

SUVI: Single-Stream Vision AI

Visually grounded multimodal AI with a smaller active inference path

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Code

github.com/nakulvyas21/suvi-arm-optimization

Checkpoint

huggingface.co/nakulvyas21/suvi-phase3-flickr30k-arm

The problem: multimodal AI carries an always-on vision tower

Most vision-language systems keep a language model and a separate vision encoder active for every image request.

Typical dual-tower path

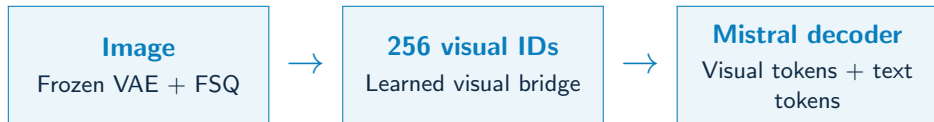
Image → vision tower → projector → language model

- More parameters in the active path
- Higher memory pressure
- More runtime complexity on constrained infrastructure

What Arm makes possible

Efficient cloud and edge deployment, if the multimodal stack is designed for a smaller resource envelope.

SUVI: image and language in one decoder stream



- SUVI means **Scalable Unified Vector Intelligence**.
- It represents visual evidence as discrete tokens in the same autoregressive stream as language.
- The result is a simpler multimodal deployment path without a separate ViT or CLIP vision tower active during cached serving.

Introduced in the IEEE CAI 2026 paper: SUVI: Scalable Unified Vector Intelligence for Efficient Edge Deployment.

First establish the architecture: visual grounding

Phase 3 is a post-paper SUVI development: it retains the discrete FSQ image representation and adds a learned visual bridge into the Mistral input stream.

Held-out Flickr30k causal-prefix test

Real image prefixes outperformed shuffled-image, fixed-prefix, and random-prefix controls.

What this establishes

- The visual path carries image-specific information.
- The learned bridge connects visual IDs to the language decoder.
- SUVI is visually grounded, rather than treating every image as the same prefix.

What comes next

- Scale training for richer caption quality.
- Extend evaluation to broader visual tasks.
- Keep the single-stream resource advantage as the system scales.

Arm deployment case study: compact caching + core scaling

1 KB FSQ visual cache

An image is represented by a compact record of visual IDs. For cached images, those IDs reconstruct the same visual prefix and caption token output as uncached processing.

Eight-core Arm scaling

Cached request time improved from 54.98 seconds at one thread to 18.12 seconds at eight threads.

1 KB

per cached image

Cached SUVI scaling

Arm cores	Median request
1	54.98 s
2	36.50 s
4	24.76 s
8	18.12 s

3.0× faster at 8 cores

The cache and the decoder form a repeatable, compact serving path for reusable images.

Deployment result: lower memory and lower latency on Arm

8-core Google Axion Arm VM. Same request conditions for the resource comparison.

Deployment path	Peak RSS	Median request
LLaVA v1.6 Mistral-7B dual tower	20.72 GB	26.01 s
SUVI uncached	15.36 GB	18.79 s
SUVI cached	15.36 GB	18.12 s

25.9%

less peak memory

30.3%

lower median latency

identical

cached visual output

Why this matters

SUVI makes a clear architectural trade

Move visual evidence into the language model's token stream, then optimize that compact stream for Arm.

- **For deployment:** lower memory and latency than the dual-tower comparator.
- **For Arm:** compact FSQ caching and effective eight-core CPU scaling.
- **For research:** a public, reproducible visual-grounding checkpoint and benchmark path.

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